University Singidunum

Applied Artificial Intelligence

Final Exam Project Instructions

Deep Learning and Artificial Neural Networks

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Overview

This document outlines the expectations, requirements, and assessment criteria for the individual final project in the course *Deep Learning and Artificial Neural Networks*. Students are expected to select a relevant topic, implement a solution, and reflect on the results in a structured report.

General Instructions

- This is an **individual project**. Each student is expected to submit independent and original work.
- The objective is to design and implement a deep learning-based solution inspired by any of the topics covered during the course.
- The project should reflect a comprehensive understanding of the chosen topic, including correct application of algorithms, data processing, model training, evaluation, and interpretation.
- Suggested topics include, but are not limited to:
  - Fundamentals of neural networks (e.g., NN for classification or regression)
  - Convolutional neural networks for image classification
  - Object detection using YOLOv5 or Faster R-CNN
  - Semantic segmentation using U-Net
  - Recurrent neural networks for sequence modeling
  - Natural language processing tasks (e.g., sentiment analysis, text classification)
  - Transformer-based models and attention mechanisms
  - Experiments on optimization, regularization, or hyperparameter tuning
- Plagiarism or identical submissions will not be tolerated and will result in a failing grade.

Deliverables

- A Jupyter Notebook (.ipynb) file containing:
  - Model implementation and training procedure
  - Dataset preparation and preprocessing steps
  - Evaluation metrics and result visualizations
  - Well-documented and commented code
- A concise report (1–2 pages in PDF format), summarizing:
  - Problem definition and motivation
  - Description of the methodology
  - Results, analysis, and concluding remarks

Grading (Maximum 30 Points)

- Clarity of problem formulation and motivation – **5 points**
- Model design and implementation quality – **10 points**
- Evaluation methodology and result interpretation – **8 points**
- Code readability and structure – **4 points**
- Report organization and critical insight – **3 points**

Submission

- Deadline: **24 hours prior to the scheduled final exam**.
- Submit the Jupyter notebook and the report via email to:
  - `tbezdan@singidunum.ac.rs` (Lecturer)
  - `avesic@singidunum.ac.rs` (Teaching Assistant)